Investigation of the Continuum Limit in the Presence of Small Breaking of Gauge Invariance in Lattice Gauge Theories

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This thesis investigates whether and how the continuum limit of a lattice gauge theory is affected by small explicit violations of gauge invariance. The study combines theoretical expectations from lattice regularization and symmetry restoration with numerical measurements of Wilson loops, Creutz ratios, the static potential, the Sommer scale, and related scaling observables. The central goal is to determine whether gauge-breaking effects vanish, persist, or modify the extrapolation toward the continuum limit as the lattice spacing is reduced.

Contents

Notation and Conventions

a Lattice spacing.

β Bare lattice coupling parameter.

ε Parameter controlling the explicit breaking of gauge invariance.

$W(R,T)$ Wilson loop with spatial extent R and temporal extent T.

$\chi(R,T)$ Creutz ratio used to estimate the force and string tension.

r_0 Sommer scale defined through $r^2F(r) = 1.65$.

σ String tension.

1. Introduction

1.1. Motivation

Explain why gauge invariance is normally central in lattice gauge theory and why controlled violations are relevant. Possible motivations include numerical robustness, effective descriptions, regulator artifacts, quantum simulation errors, or imperfect implementation of gauge constraints.

1.2. Research Question

State the main question precisely:

> Does a small explicit breaking of gauge invariance disappear in the continuum limit, or does it alter the continuum extrapolation of physical observables?

1.3. Strategy and Main Observables

Introduce the numerical strategy and the observables used to probe the continuum limit: Wilson loops, Creutz ratios, the static potential $V(R)$, r_0, and σ.

1.4. Thesis Outline

Summarize the role of each chapter.

2. Theoretical Background

2.1. Gauge Theories in the Continuum

2.1.1. Gauge Fields and Local Gauge Symmetry

2.1.2. Gauge-Invariant Observables

2.1.3. Physical Consequences of Gauge Invariance

2.2. Lattice Gauge Theory

2.2.1. Discretization of Space-Time

2.2.2. Link Variables and Plaquettes

2.2.3. Wilson Gauge Action

2.2.4. Path Integral and Expectation Values

2.2.5. Continuum Limit and Renormalization

2.3. Static Potential and Confinement

2.3.1. Wilson Loops

2.3.2. Effective Potential Extraction

2.3.3. Creutz Ratios

2.3.4. Sommer Scale and String Tension

3. Small Breaking of Gauge Invariance

3.1. Definition of the Gauge-Breaking Deformation

Define the modified lattice action or update rule used in the simulations. Specify the breaking parameters, for example ε_1 and ε_2, and explain which gauge transformations are no longer exact symmetries.

3.2. Symmetry and Scaling Expectations

3.2.1. Relevant, Marginal, and Irrelevant Breaking Terms

3.2.2. Possible Restoration of Gauge Invariance

3.2.3. Expected Dependence on ε and a

3.3. Working Hypotheses

List the concrete hypotheses tested numerically:

1. Gauge-invariant observables approach the same continuum values for sufficiently small ε.
2. Gauge-breaking effects scale to zero with a measurable power of a or ε.
3. Above a threshold in ε, continuum extrapolations become unstable or incompatible with the gauge-invariant limit.

4. Numerical Setup

4.1. Simulation Parameters

Describe the gauge group, lattice volumes, coupling range, update algorithm, thermalization strategy, number of sweeps, and stored observables.

4.2. Ensemble Organization

Explain how ensembles are grouped by common physical and algorithmic parameters. State which parameters are varied in the continuum study and which are held fixed.

4.3. Thermalization and Autocorrelations

4.3.1. Thermalization Cuts

4.3.2. Integrated Autocorrelation Time

4.3.3. Bootstrap Blocking

4.4. Analysis Pipeline

Describe the analysis code and reproducibility workflow. Include the path from raw measurements to final observables:

1. Load simulation metadata from `input.yaml`.
2. Read Wilson-loop and observable measurements.
3. Apply thermalization cuts.
4. Estimate uncertainties with bootstrap resampling.
5. Extract $V(R)$, $\chi(R,T)$, r_0, and related quantities.
6. Aggregate ensembles for continuum and breaking-strength comparisons.

5. Extraction of Physical Observables

5.1. Wilson Loop Analysis

5.1.1. Effective Mass Plateaus

5.1.2. Choice of Fit Windows

5.1.3. Systematic Uncertainty from Fit Ranges

5.2. Static Potential

5.2.1. Potential Extraction

5.2.2. Cornell-Type Fits

5.2.3. Force and Sommer Scale

5.3. Creutz Ratio Analysis

5.3.1. Definition and Interpretation

5.3.2. Large-T Behavior

5.3.3. Relation to the String Tension

5.4. Scale Setting

5.4.1. Determination of r_0/a

5.4.2. Alternative Scale Estimates

5.4.3. Consistency Checks

6. Continuum Extrapolation Method

6.1. Choice of Continuum Variable

Define the extrapolation variable, for example $(a/r_0)^2$, $1/\beta$, or another measured scale proxy.

6.2. Fixed-Physics Trajectories

Explain how ensembles at different β and ε are compared. State whether ε is held fixed in bare units, scaled with a, or tuned along the trajectory.

6.3. Fit Ansatz

6.3.1. Gauge-Invariant Baseline

6.3.2. Linear and Quadratic Lattice-Spacing Corrections

6.3.3. Combined Fits in a and ε

6.4. Uncertainty Budget

6.4.1. Statistical Errors

6.4.2. Fit-Window Systematics

6.4.3. Finite-Volume Effects

6.4.4. Continuum-Fit Systematics

7. Results

7.1. Thermalization and Data Quality

Insert representative thermalization histories and autocorrelation estimates.

7.2. Gauge-Invariant Baseline

Present the $\varepsilon = 0$ results for $W(R,T)$, $V(R)$, r_0, and σ. This establishes the reference continuum behavior.

7.3. Dependence on Gauge-Breaking Strength

7.3.1. Observable Shifts at Fixed β

7.3.2. Scaling of Shifts with ε

7.3.3. Thresholds or Nonlinear Behavior

7.4. Continuum Limit at Fixed Breaking

7.4.1. Continuum Extrapolations for Each ε

7.4.2. Comparison with the Gauge-Invariant Limit

7.4.3. Stability Under Fit Variations

7.5. Combined Scaling Analysis

Fit the dependence on both lattice spacing and gauge-breaking parameter. State whether the data support restoration of gauge invariance in the continuum limit.

7.6. Finite-Volume and Systematic Checks

Collect checks that are not part of the primary result but constrain the interpretation.

8. Discussion

8.1. Interpretation of the Continuum Limit

Discuss whether small gauge breaking behaves like an irrelevant lattice artifact, a relevant deformation, or a source of uncontrolled continuum bias.

8.2. Comparison with Theoretical Expectations

Relate the numerical findings to the hypotheses from Chapter 3.

8.3. Limitations

Discuss limits from statistics, accessible lattice spacings, finite volume, fit-systematic control, and the specific form of gauge breaking studied.

8.4. Implications

Explain what the result means for simulations or formulations where gauge invariance is only approximate.

9. Conclusion and Outlook

9.1. Summary of Findings

State the central quantitative conclusions.

9.2. Answer to the Research Question

Give the direct answer to whether the continuum limit is robust under the investigated small breaking of gauge invariance.

9.3. Future Work

Possible extensions:

- Study additional forms of gauge-breaking terms.
- Increase statistics and extend to finer lattice spacings.
- Add gauge-variant diagnostics to directly measure symmetry restoration.
- Compare different gauge groups or dimensions.

A. Simulation Tables

Provide complete tables of ensembles, including β, lattice size, breaking parameters, number of sweeps, thermalization cuts, and measured scales.

B. Additional Plots

Include supplementary Wilson-loop plateaus, bootstrap block-size scans, potential fits, and continuum-fit variants.

C. Analysis Code and Reproducibility

Document the exact command-line workflow, cache version, software environment, and data-selection rules used for the final analysis.

D. Derivations

Collect longer derivations that would interrupt the main text, such as Creutz-ratio identities or continuum-scaling arguments.